

No. of Printed Pages : 4
Roll No.

221043

4th Sem / ECE, ECE(For Speech and Hearing Impaired)

Subject : Communication Systems

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Frequency range of VHF band is

- a) 3MHz to 30 MHz
- b) 300 MHz to 3000 MHz
- c) 30 MHz to 300 MHz
- d) 30 KHz to 300 kHz

Q.2 Height of geostationary orbit is

- a) 3600 km
- b) 18000 km
- c) 54000 km
- d) 36000 km

Q.3 One Mhz is equal to

- a) 10^6 Hz
- b) 10^9 Hz
- c) 10^3 Hz
- d) 10^{12} Hz

Q.4 If N_m is the maximum electron density then critical frequency is given by

- a) $9 N_m$ b) $9 \sqrt{N_m}$
c) $9/\sqrt{N_m}$ d) $2 \sqrt{N_m}$

Q.5 Which type of antenna is having driven element, directors and reflectors.

- a) Yagi-Uda b) Loop
c) Ferrite rod. d) Patch

Q.6 Modulation is done in _____

- a) Radio Receiver b) Antenna
c) If Amplifier d) Transmitter

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Full form of IF is _____

Q.8 What is the full form of MUF?

Q.9 VHF stands for _____

Q.10 The electric and magnetic fields are _____ to each other in EM waves.

Q.11 Define Antenna.

Q.12 Define Geo-stationary satellite.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain block diagram of AM Transmitter.

Q.14 Classify transmitter on the basis of modulation?

Q.15 Explain fidelity and S/N ratio.

Q.16 Explain delayed AGC.

Q.17 Discuss half wave dipole antenna.

Q.18 Explain Yagi-Uda antenna.

Q.19 What is ground wave propagation.

Q.20 Explain the terms

a) MUF

b) Virtual height

Q.21 Explain Ionosphere & its layers?

Q.22 Explain polarization of electromagnetic waves.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the working of super heterodyne receiver with block diagram.

Q.24 Explain electromagnetic spectrum and its various ranges.

Q.25 Explain the following

- a) line of sight propagation
- b) Reactance FET transmitter